

is typically used, partly from practical considerations such as making and handling thin wafers with ease and for surface passivation reasons.

The front side of a PV cell is textured with a few micrometres of standing, round-shaped valleys, as shown in Fig. 6, to increase the amount of light coupled to the cell. Functions of the texture are to increase transmis-

sion of incoming photons into the silicon absorber and to increase the path length of photons inside the absorber.

As shown in Fig. 5, the front side is further coated with 75-nanometre-thick hydrogenated silicon nitride with a refractive index of 2.1, enabling reduction of front reflection. Fig. 5 also shows that, at the front side, front contact of continuous H-like pattern of a thin transparent silver conductor is formed from sintered silver paste that penetrates into silicon to a depth of thousand nanometres for efficient carrier transport and connected to busbars for collection of current.

The front-side top layer is doped with phosphorus to form a p-n junction, enabling diffusion of a minority carrier (hole) within the layer and to allow sufficient conduction to enable efficient transport of majority carriers (electrons) to the contact. The rear side is fully metallised with aluminium, alloyed with silicon, with contact pads connected to the busbar for efficient carrier transport. The alloying process produces a layer of silicon heavily doped with aluminium, which reduces the recombination rate at the rear end.

### The PV-MD process

Recently, scientists have fabricated a PV-MD device that can stably produce

more than 1.60 kilograms of clean water per hour per square-metre of PV surface from sea water, while simultaneously generating electricity with an efficiency of more than eleven per cent under one Sun irradiation. Its high clean water production rate is realised by constructing a multi-stage membrane distillation (MSMD) device at the back of the solar cell to recycle the latent heat of water vapour condensation at each distillation stage. This composite device can significantly reduce capital investment costs by sharing the same land and the same mounting system.

Membrane distillation is a thermal separation process that can be used for extraction of clean water. Mass transfer occurs through a porous membrane, which cannot be wetted for the solvent. Two liquids with different temperatures contact the membrane. On input side (feed side), it is the warm aqueous solution, for example, sea water; and on effluent side (distillate side), it is condensed steam diffused through the membrane.

Feed temperatures in MD typically range from 40°C to 85°C. The driving force in MD is the partial pressure difference between each side of the membrane pores. Temperature difference between the two liquids leads to vapour pressure difference across the membrane. Due to the hydrophobic nature of the membrane, only vapour can pass across the membrane, and not any liquid solution being distilled (Fig. 7).

Membranes should be hydrophobic and micro-spongy. These are mostly produced from polytetrafluorethylene (PTFE), polypropylene (PP) or polyvinylidene-fluoride (PVDF). Pores must be large enough to allow mass flow, but small enough to prevent intrusion of water under operating conditions. Membranes with pore diameter of 0.01mm to 1.00mm can be used for membrane distillation. Whenever salts are accumulated on the membranes, as shown in Fig. 8, the same should be removed by washing.

MD is a separation process that could operate with minimum external energy requirements and least capital and land for the plant. Required equipment for this process are much

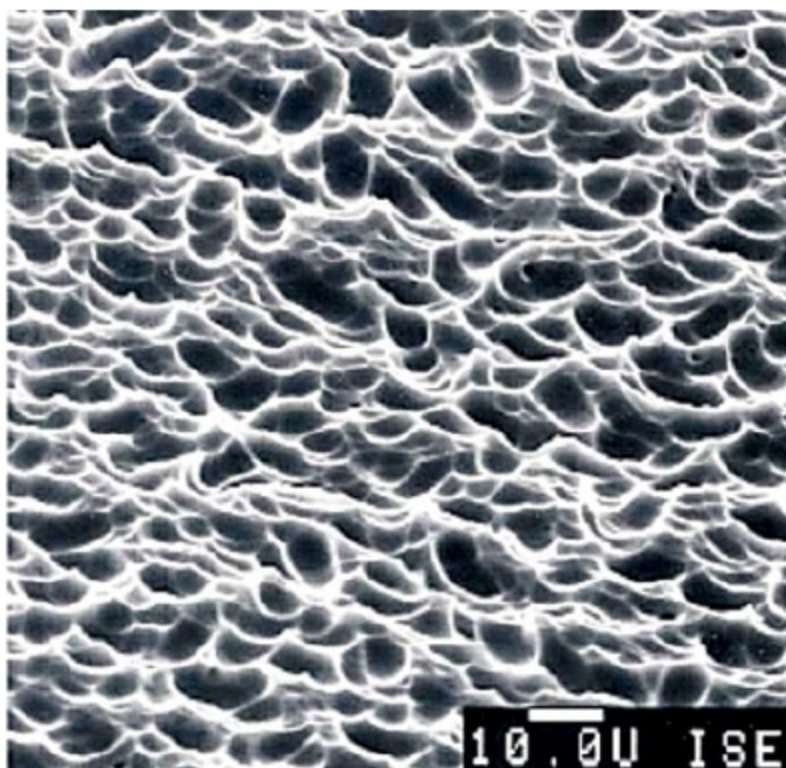


Fig. 6: Texture of top surface of polycrystalline solar cell

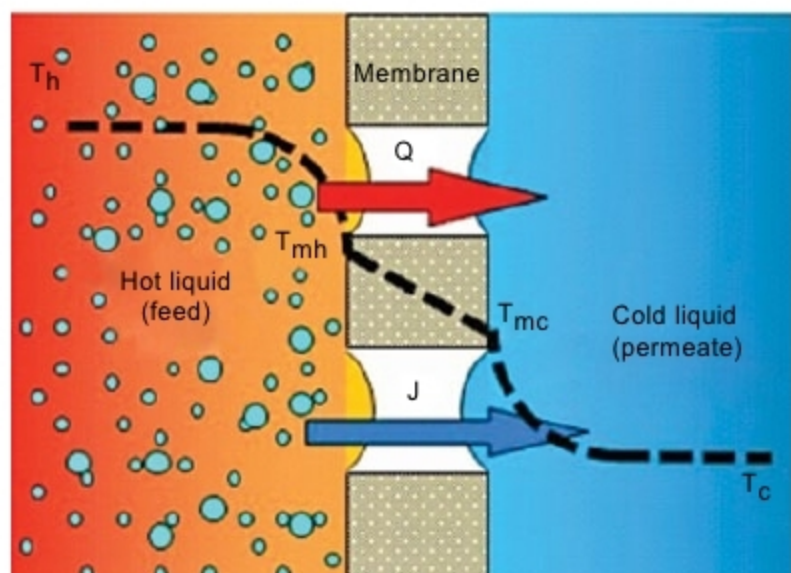


Fig. 7: Basic principle of membrane distillation process

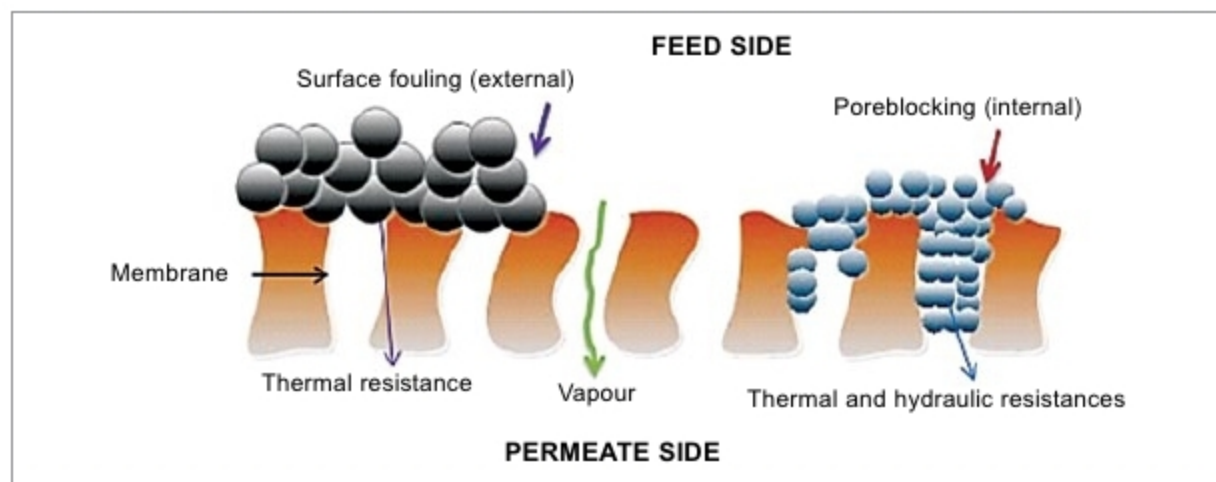


Fig. 8: Mechanism of fouling in membrane